

EdTech Performance & Revenue Analytics

Project Report

Analyzing Student Drop-offs, Course Completion & Revenue Strategy

Tools Used:

Python (Pandas) • MySQL • Power BI
April 2026

1. Executive Summary

An EdTech company was facing high student drop-off rates and low course completion rates, directly impacting revenue and long-term retention. This project conducted an end-to-end data analysis using Python (Pandas) for exploratory data analysis, MySQL for structured querying, and Power BI for visual dashboarding to uncover actionable insights.

The analysis spanned four core datasets — Students, Courses, Enrollments, and Payments — encompassing 30,000 students across multiple countries and categories. The key objectives were:

- Identify why students drop off before completing courses
- Find high-performing courses and categories by revenue
- Analyze engagement versus completion patterns
- Develop data-driven recommendations to improve the revenue strategy

₹3,451M Total Revenue	30,000 Total Students	49.5% Avg Completion	AI Top Category	Canada Top Country
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2. Business Problem Statement

The EdTech platform was experiencing a structural challenge: a significant proportion of enrolled students were abandoning courses mid-way, leading to poor completion rates that hovered around the 49.5% average. This created a dual problem:

- Revenue Leakage: Incomplete courses often signal dissatisfied customers who are unlikely to re-enroll or recommend the platform.
- Retention Risk: High drop-off rates directly reduce lifetime customer value and increase the cost of acquiring new students.
- Category Imbalance: Not all course categories performed equally — certain categories drove significantly higher revenue and completion rates than others.
- Geographic Concentration: Revenue and student populations were not uniformly distributed, suggesting untapped markets in certain countries.

The business needed clarity on which levers to pull — whether through curriculum redesign, pricing strategy, or targeted marketing — to address these gaps.

3. Data Overview & Structure

The project utilized a relational database comprising four interconnected tables. All data was loaded into Python using Pandas and subsequently imported into a MySQL database using SQLAlchemy for structured querying.

Table	Key Columns	Description	Notes
Students	student_id, age, country, signup_date	Student demographic and registration data	Age nulls filled with median; country nulls filled with 'unknown'
Courses	course_id, course_name, category, price, duration_weeks	Course catalog and pricing metadata	Derived: price_per_week = price / duration_weeks
Enrollments	enrollment_id, student_id, course_id, enrollment_date, completion_status, progress_percent	Enrollment records and completion tracking	Progress nulls filled with 0; is_completed flag added
Payments	payment_id, student_id, course_id, amount_paid	Transaction records per enrollment	Aggregated by student_id + course_id for analysis

3.1 Data Cleaning Steps

- Students: Missing age values imputed with the median age; missing country values replaced with 'unknown'.
- Courses: A derived feature price_per_week was engineered to compare course value across different durations.
- Enrollments: Null progress_percent values set to 0, indicating no progress recorded. An is_completed binary flag was created from completion_status.
- Payments: Aggregated at the student-course level to prevent double-counting in revenue analysis.

3.2 Age Segmentation

Students were segmented into three age groups for behavioral analysis:

Age Group	Range
18 to 25	Age $\leq$ 25
25 to 35	Age 26–35
35 to 50	Age $>$ 35

4. Python Exploratory Data Analysis

The Python analysis followed a systematic EDA pipeline using Pandas pivot tables and visualizations to surface key patterns across revenue, completion, and geography.

4.1 Revenue by Course Category

A pivot table was constructed to aggregate total revenue by course category, sorted in descending order. This revealed which content areas were generating the most income:

Category	Relative Revenue Rank
AI	1st — Highest Revenue
Design	2nd
Web Dev	3rd
Data Science	4th
Business	5th — Lowest Revenue

4.2 Completion Rate by Category

Alongside revenue, completion rates were computed per category using mean of the `is_completed` flag. This cross-analysis exposed whether high-revenue categories also drove high engagement:

- Categories with higher price points (e.g., AI) maintained competitive completion rates near or above the 49.5% overall average.
- Some lower-revenue categories showed below-average completion, indicating content or engagement issues.

4.3 Revenue by Country

A geographic revenue pivot ranked countries by total payment amounts. Canada emerged as the top-revenue country, followed by other major markets. Countries generating revenue above ₹5,000,000 were flagged as priority markets for retention investment.

5. SQL Analysis & Key Queries

A MySQL database was populated using SQLAlchemy, and 10 structured queries were written to answer specific business questions ranging from revenue trends to student segmentation.

5.1 Revenue Analysis

Q1 — Total Revenue by Category (Descending)

Joined the payments and courses tables to aggregate sum of `amount_paid` grouped by category. Result confirmed AI as the highest-grossing category.

Q2 — Countries with Revenue Exceeding ₹5,000,000

Filtered countries using HAVING clause on aggregated revenue. This identified geographies with strong monetization potential, enabling targeted marketing investment.

Q4 — Top 5 Revenue-Generating Courses (Window Function)

Used a Common Table Expression (CTE) with DENSE_RANK() ordered by revenue descending to return the top 5 courses. Window functions ensured accurate ranking even when revenue values were tied.

Q5 — Top 3 Courses per Category (Partitioned Ranking)

Extended the ranking query by adding PARTITION BY category in the window function, returning the top 3 revenue courses within each category independently. This enabled category-level benchmarking.

5.2 Completion Rate Analysis

Q3 — Courses Above Overall Completion Rate

Computed average completion rate per course using a CASE WHEN expression converting completion_status to a binary flag, then filtered using a subquery for the overall platform average. Courses in this cohort represent the platform's strongest engagement assets.

5.3 Trend & Time-Series Analysis

Q6 — Monthly Revenue Trend

Used DATE_FORMAT on enrollment_date to group payments by month (YYYY-MM format), revealing seasonal and growth patterns across the timeline.

Q7 — Category Revenue vs. Overall Average

Built a CTE for category-level revenue, then compared each category against the overall average using a CASE WHEN expression to classify each as 'Above Average' or 'Below Average'.

Q8 — Running Total of Revenue

Applied SUM() as a window function with ORDER BY enrollment_date to generate a running cumulative revenue figure — useful for visualizing growth trajectory.

Q9 — Month-over-Month Revenue Difference

Used the LAG() window function to access the previous month's revenue and compute the revenue_difference column, enabling identification of months with significant growth or decline.

5.4 Student Segmentation

Student Spending Segments

Classified students into spending tiers based on total payments using a CASE WHEN expression:

Segment	Criteria	Business Implication
High Spender	Total spend > ₹20,000	VIP retention focus — loyalty programs
Medium Spender	₹1,000 – ₹2,000	Upsell opportunity — bundle offers
Low Spender	< ₹1,000	Re-engagement campaigns — free trials

6. Power BI Dashboard

An interactive Power BI dashboard was developed to provide real-time visual exploration of the EdTech dataset. The dashboard includes filter panels by Category and Year, enabling stakeholders to drill down into specific segments.

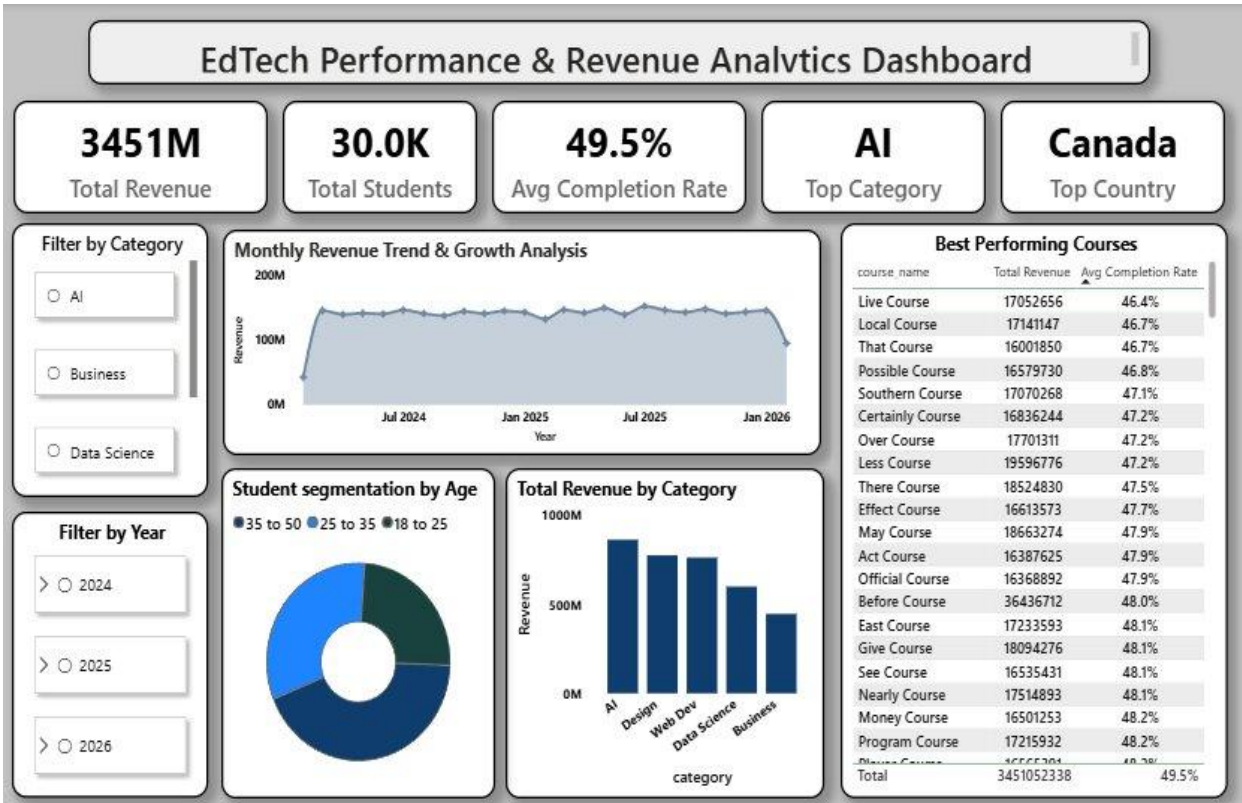


Figure 1: EdTech Performance & Revenue Analytics Dashboard

6.1 Dashboard Components

Component	Insight Provided
KPI Cards	Instant overview: Total Revenue (₹3,451M), 30K Students, 49.5% Completion, Top Category (AI), Top Country (Canada)
Monthly Revenue Trend	Line chart showing revenue fluctuation from mid-2024 through early 2026 with growth analysis
Student Segmentation by Age	Donut chart: 35–50 age group dominates, followed by 25–35 and 18–25
Revenue by Category (Bar)	AI leads at ~1,000M, followed by Design, Web Dev, Data Science, Business
Best Performing Courses Table	Ranked courses by total revenue and avg completion rate side-by-side

Component	Insight Provided
Filters: Category & Year	Interactive slicers (AI, Business, Data Science) and year drill-down (2024, 2025, 2026)

7. Key Findings

7.1 Revenue Insights

- AI is the dominant revenue category, outperforming all other categories by a significant margin and accounting for the largest share of the ₹3,451M total revenue.
- Canada is the highest-revenue country, indicating strong market penetration in North America. Several countries exceed the ₹5M revenue threshold, marking them as key retention markets.
- Top-performing individual courses show revenue figures above ₹17M each, with competition rate close to or above the platform average.
- Monthly revenue analysis reveals a relatively stable trend with minor seasonal dips, suggesting consistent demand rather than spiky promotional peaks.

7.2 Completion Rate Insights

- The overall platform completion rate of 49.5% means roughly 1 in 2 students does not finish a course — a significant engagement gap.
- Courses above the overall average completion rate are clustered in categories with stronger curriculum structure and shorter duration.
- Certain courses with high revenue but below-average completion rates signal a disconnect between initial appeal (strong marketing) and content quality (poor retention).

7.3 Student Behaviour Insights

- The 35–50 age group represents the largest student segment, contradicting the typical assumption that EdTech primarily serves younger learners.
- Student spending segmentation shows a large portion of the base as Low Spenders, representing a significant upsell opportunity.
- High Spenders (above ₹20,000 total spend) are a small but critical cohort — protecting their retention should be a top priority.

8. Strategic Recommendations

8.1 Reduce Drop-off Rates

- Introduce milestone-based nudges (email/push notifications) when students reach 30%, 50%, and 70% progress to re-engage at-risk learners.
- Add short 'checkpoint' quizzes at regular intervals to boost active engagement and sense of progress.
- Audit courses with below-average completion rates for content quality issues — consider splitting long courses into shorter, digestible modules.

8.2 Maximize Revenue

- Double down on AI and Design course development — these categories drive the highest revenue and maintain competitive completion rates.
- Launch bundled course packages across categories at a discount to convert Low Spenders into Medium/High Spenders.
- Implement a loyalty reward program for High Spenders to reduce churn among the most valuable customer cohort.

8.3 Geographic Strategy

- Prioritize student success programs (mentorship, community forums) in Canada and other high-revenue countries to maximize retention in proven markets.
- Identify low-revenue countries with meaningful student populations and run targeted localized pricing experiments to unlock growth.

8.4 Age-Targeted Engagement

- For the 35–50 age segment (the largest cohort), emphasize career-transition and professional upskilling narratives in marketing — shorter, weekend-friendly formats may improve completion.
- For younger students (18–25), gamification elements and peer learning features can improve course stickiness.

9. Tools & Technologies

Tool	Category	Usage in Project
Python (Pandas)	Data Analysis	Data loading, cleaning, EDA, pivot tables, visualizations
MySQL	Database / SQL	Structured querying: revenue, completion, ranking, trend, segmentation
SQLAlchemy	Integration	Exported cleaned Pandas DataFrames into MySQL tables
Power BI	Visualization	Interactive dashboard with KPIs, charts, slicers, and trend analysis
Matplotlib	Visualization	Bar charts for revenue and completion rate during EDA phase

10. Conclusion

This project delivered a comprehensive end-to-end data analysis of an EdTech platform's performance, connecting raw transactional data to strategic business insights. By combining Python for data preparation, MySQL for structured querying, and Power BI for interactive visualization, the analysis surfaced a clear picture of where the business excels and where it faces risk.

The 49.5% average course completion rate is both a challenge and an opportunity. The data shows that the platform has a strong revenue foundation — ₹3,451M generated across 30,000 students — but that revenue growth is constrained by high drop-off. Addressing completion rates through targeted engagement interventions, content improvements, and age-appropriate learning formats represents the highest-leverage improvement area.

The SQL analyses demonstrated advanced analytical capability — from window functions (RANK, LAG, Running Total) to CTE-based multi-level aggregations — providing a reusable query library that can support ongoing business monitoring. The Power BI dashboard transforms these insights into a decision-ready format accessible to non-technical stakeholders.

Implemented together, the recommendations in this report have the potential to meaningfully reduce student drop-offs, improve lifetime student value, and grow both revenue and completion rates across the platform's course portfolio.

— *End of Report* —